

Dummy Load for Amplifier Testing

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If you want to test an audio amplifier, here is a compact fan-cooled dummy load that can be used to provide any of the three commonly needed load impedances (4 Ω , 6 Ω or 8 Ω), with power rating varying from 125 W to 250 W. Interestingly, this dummy load can be made using readily avail-

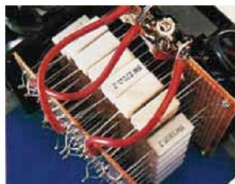


Fig. 1: Author's prototype

able and relatively low-cost parts.

If you ever need to check the performance of an audio amplifier, a resistive dummy load can be a very handy tool. For accurate testing of aspects such as frequency performance, power output and signal-to-noise ratio, the 'flat' performance of a resistive dummy load is virtually essential as even the best speakers tend to have peaks and troughs in their impedance curve, which can distort your measurements.

In any case, you wouldn't want to have speakers connected to an amplifier undergoing power output tests. Even if the speakers could cope with the continuous power output without damage, the resulting sound levels would be deafening.

To make the dummy load useful for testing modern amplifiers, it is necessary to provide a choice of at



least two load impedances (8 Ω and 4 Ω), and preferably 6 Ω as well, which can cope with power levels comfortably above 100 W for the 8 Ω load and somewhat higher for the two lower impedances. A small (80mm) computer-type 'muffin' cooling fan is used to help the resistor cope with dissipating high powers.

Circuit and working

Fig. 2 shows the arrangement of resistor banks and Fig. 3 shows the complete circuit. The circuit is built around 60 resistors (R1 through R60). The load resistors are connected in three parallel groups. Two groups with 25 resistors each form two sets of 8 Ω , 125W loads. The third group forms the load of 24 Ω , 50 W.

One of the 8 Ω groups is connected permanently across output terminal CON1 and forms a basic 8 Ω load, while the other two groups are switched in parallel with the existing 8 Ω load by switch S1, whenever required, to provide either 6 Ω or 4 Ω load impedance.

When switch S1 is at position 3, only the 8 Ω group is connected across CON1, and forms the basic 8 Ω load. When switch S1 is at position 1, 4 Ω is connected across the load output terminals. 6 Ω load can be selected by placing switch S1 in position 2.

The accuracy of the actual impedance levels provided by the final dummy load is determined mainly

Test Points

Test point	Details
Across TP0 and TP1	8 Ω , 125W when S1 is at position 3
Across TP0 and TP2	6 Ω , 175W when S1 is at position 2
Across TP0 and TP3	4 Ω , 250W when S1 is at position 1

PARTS LIST

<i>Resistors (all wire-wound):</i>	
R1-R12,	- 180 Ω , 5W
R36-R47,	- 220 Ω , 5W
R13-R29,	- 270 Ω , 5W
R48-R60	- 220 Ω , 5W
R30-R35	- 270 Ω , 5W
<i>Miscellaneous:</i>	
S1	- 3-way rotary switch
CON1	- 2-pin connector

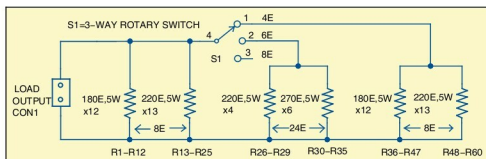


Fig. 2: Resistor banks arrangement

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